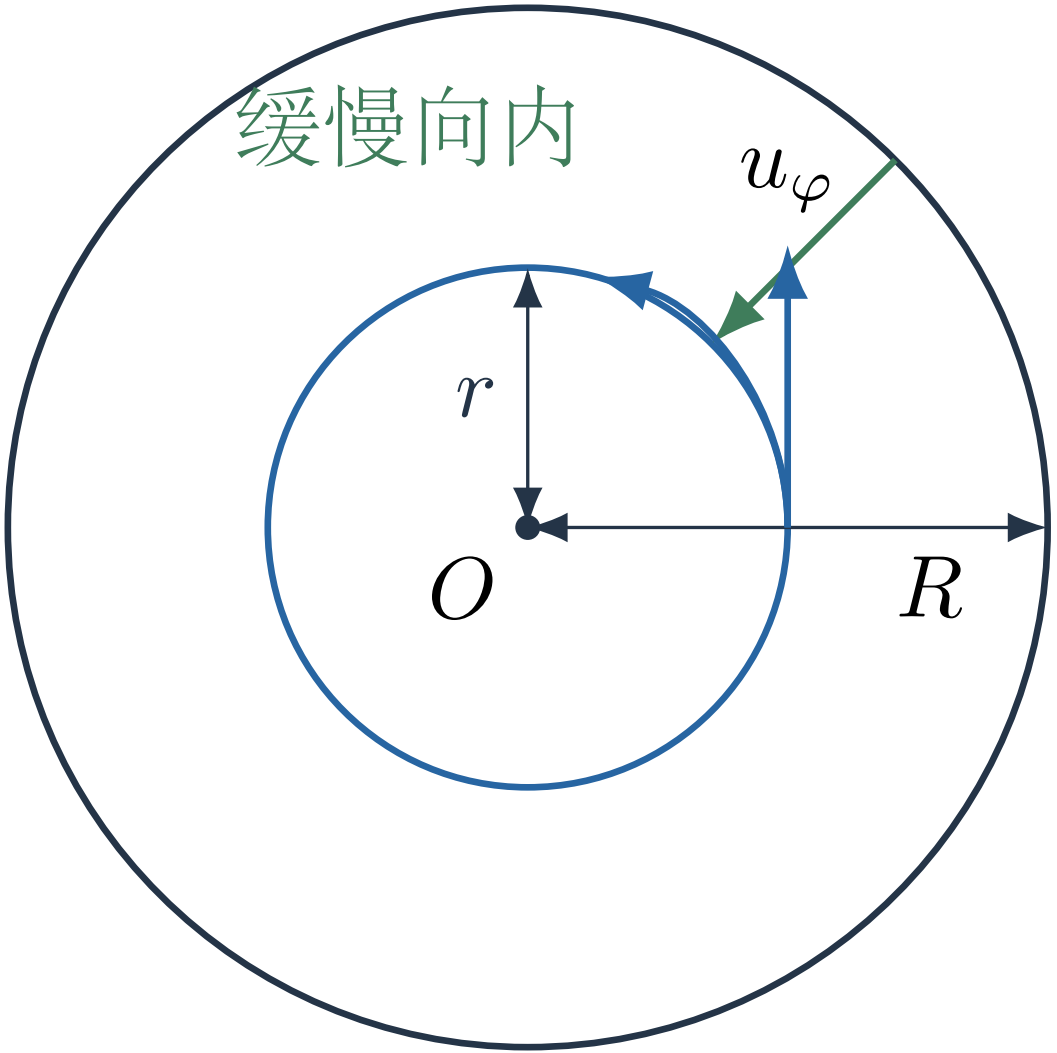

$$2\pi R u_\varphi(R) = \Gamma_0$$

$$2\pi r u_\varphi(r) = \Gamma_0$$